

**Zewail City of Science and Technology**

Advanced RF/Mixed Signal ICs | NANENG 522 | Spring 2026

**Low Noise Amplifier Design**

65 nm CMOS Technology | 5 GHz | Narrowband

**Phase 3 Report**

Post-Layout Simulation, Physical Design & Final Discussion

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**Submitted to: Dr. Nehad Moustafa**

| Name           | Student ID |
|----------------|------------|
| Ahmed Fouda    | 202101219  |
| Maher Hashem   | 202200178  |
| Adham Mohamed  | 202200322  |
| Ahmed Atef     | 202200544  |
| Mohamed Ahmed  | 202301194  |
| Ahmed Mohammed | 202301745  |

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# 1. Introduction & Design Target

This Phase 3 report presents the complete finalisation of the 5 GHz narrowband Low Noise Amplifier (LNA) designed in TSMC 65 nm CMOS technology for the IEEE 802.11a wireless standard. Building directly on the schematic-level simulation results of Phase 2, this report covers: the final design methodology and hand-analysis framework including the novel linearisation technique, updated component values, DC operating-point confirmation, on-chip spiral inductor characterisation, pre-layout RF performance (gain, noise figure, S-parameters, IIP3, stability), post-layout parasitic-extraction (PEX) simulation results, and a comprehensive final discussion analysing the schematic-to-layout performance degradation.

The amplifier topology is a Cascode Common-Source with Inductive Source Degeneration (CS-ISD), a well-established architecture for narrowband LNA design in CMOS that simultaneously achieves low noise figure, high gain, and unconditional stability. A key innovation in this design is the addition of an external capacitor  $C_{ext} = 240$  fF in parallel with the intrinsic gate-source capacitance  $C_{gs}$  of the input transistor M0. This modification was introduced to resolve a linearity deficit observed during Phase 2 optimisation, and its theoretical justification is developed in detail in Section 2.

## 1.1 Target Specifications

Table 1 restates the complete set of design specifications against which all simulation results are evaluated throughout this report.

Table 1: LNA Design Specifications

| Parameter                      | Specification | Notes                   | Remarks                  |
|--------------------------------|---------------|-------------------------|--------------------------|
| Gain ( $S_{21}$ )              | > 25 dB       | Small-signal power gain | Target: IEEE 802.11a     |
| Noise Figure (NF)              | < 2 dB        | At 5 GHz                | Critical for sensitivity |
| Input Return Loss ( $S_{11}$ ) | < -12 dB      | Input match quality     | 50 $\Omega$ source       |
| Drain Current ( $I_{DD}$ )     | < 5 mA        | RF core only            | Excl. biasing branch     |
| IIP3                           | > -5 dBm      | Two-tone test           | 10 MHz tone separation   |
| Center Frequency               | 5 GHz         | IEEE 802.11a band       | —                        |

|                       |        |                 |                     |
|-----------------------|--------|-----------------|---------------------|
| Supply Voltage (V_DD) | 1.8 V  | Nominal         | Simulator: 1.2 V    |
| Output Load (C_Load)  | 500 fF | Fixed load      | Spec requirement    |
| Input Pad Cap (C_pad) | 50 fF  | Each pin to GND | ESD/bond-wire model |

## 2. Theoretical Analysis & Hand Calculations

### 2.1 Design Methodology Overview

The design proceeded through a structured, five-stage hand-analysis methodology. Rather than selecting the transistor width arbitrarily for maximum gain, we adopted a power-constrained noise optimisation (PCNO) approach where each stage informed the next, ultimately leading to the need for the  $C_{ext}$  linearisation technique.

#### Stage 1: Device Characterisation for Minimum Noise Figure

The design begins with characterisation of the `nmos_rf` device at the fixed minimum channel length  $L_{min} = 60$  nm. For a MOSFET at RF frequencies, the minimum achievable noise figure ( $NF_{min}$ ) is given by the Friis expression for the transistor alone:

$$NF_{min} \approx 1 + 2 \cdot \left(\frac{\omega^0}{\omega_T}\right) \cdot \sqrt{\gamma \cdot \delta \cdot (1 - |c|^2)}$$

where  $\gamma \approx 2/3$  is the channel thermal noise coefficient,  $\delta \approx 4/3$  is the induced gate noise coefficient,  $c$  is the correlation coefficient between these noise sources ( $|c| \approx 0.395$ ), and  $\omega_T = g_m / C_{gs}$  is the transit angular frequency. The device was biased in the saturation region and the DC bias current was swept to find the  $V_{gs}$  that minimises  $NF_{min}$ . The key insight is that  $NF_{min}$  decreases as  $f_T$  increases (i.e., as  $g_m/C_{gs}$  increases), but increasing  $V_{gs}$  beyond the optimal point yields diminishing returns while increasing current consumption. The operating bias point was therefore selected at the  $V_{gs}$  that achieves the minimum NF for the given  $L = 60$  nm, which corresponds approximately to  $V_{gs} \approx 560$ – $570$  mV as confirmed by the DC simulation results ( $V_{gs}(M0) = 568.1$  mV).

#### Stage 2: Transistor Finger Count Selection & Current Limit

With the bias point fixed, the number of transistor fingers determines the total transistor width  $W_{total} = n_f \times w_f$  and hence the total transconductance  $g_m = g_{m\_per\_unit} \times W_{total}$ . The optimum transistor width for minimum NF under a given source impedance  $R_S = 50 \Omega$  is given by the PCNO result:

$$W_{opt} \approx \frac{1}{3 \cdot \omega^0 \cdot L_{min} \cdot C_{ox} \cdot R_S}$$

Evaluating this for the 65 nm process parameters ( $C_{ox} \approx 17$  fF/ $\mu m^2$ ,  $\omega_0 = 2\pi \times 5$  GHz,  $R_S = 50 \Omega$ ) yields  $W_{opt} \approx 60$ – $100 \mu m$ . However, this large width draws a drain current significantly exceeding the 5 mA specification. To resolve this conflict between noise optimisation and power constraint, the finger count was reduced to  $n_f = 8$  fingers at  $w_f = 1.25 \mu m$  each, giving  $W_{total} = 10 \mu m$ . This compact multi-finger layout minimises the gate resistance  $R_g$  (which contributes thermal noise)

while keeping  $I_{DD}$  well within budget. The trade-off accepted is a slight departure from the theoretical  $NF_{min}$ , but the simulated  $NF = 0.475$  dB (schematic) still far exceeds the  $< 2$  dB specification. Both M0 and M1 share identical sizing ( $W = 10$   $\mu\text{m}$ ,  $n_f = 8$ ,  $L = 60$  nm) for matching and layout symmetry. The RF NMOS model `nmos_rf` from the TSMC 65 nm PDK was selected to ensure accurate high-frequency parasitic modelling, including distributed gate resistance, substrate coupling, and velocity-saturation effects.

### Stage 3: Input Matching Network Design ( $L_s$ , $L_g$ , and $C_{ext}$ )

The CS-ISD topology synthesises a purely real  $50\ \Omega$  input impedance at resonance without any resistive element (which would add noise). The analysis proceeds as follows.

The input impedance of the inductively degenerated common-source stage, including the external capacitor  $C_{ext}$  in parallel with  $C_{gs}$ , is:

$$Z_{in(\omega)} = j\omega L_g + \frac{1}{[j\omega(C_{gs} + C_{ext})]} + j\omega L_s + g_m \cdot \frac{L_s}{C_{gs} + C_{ext}}$$

Define the total effective input capacitance as  $C_{eff} = C_{gs} + C_{ext}$ . The real part of  $Z_{in}$  at resonance is:

$$\text{Re}\{Z_{in}\} = g_m \cdot \frac{L_s}{C_{eff}} = \omega T_{eff} \cdot L_s$$

where  $\omega T_{eff} = g_m / C_{eff}$  is the effective transit frequency after adding  $C_{ext}$ . For  $\text{Re}\{Z_{in}\} = 50\ \Omega$ :

$$L_s = 50 \cdot \frac{C_{eff}}{g_m} = \frac{50}{\omega T_{eff}}$$

The imaginary part of  $Z_{in}$  must be cancelled at  $\omega_0 = 2\pi \times 5$  GHz. Setting  $\text{Im}\{Z_{in}\} = 0$ :

$$\omega^0(L_g + L_s) - \frac{1}{\omega^0 \cdot C_{eff}} = 0$$

$$L_g = \frac{1}{\omega^0 \cdot C_{eff}} - L_s$$

From the Cadence simulation with  $C_{eff} = C_{gs} + C_{ext}$  ( $C_{gs}$  from the 8-finger device at the chosen bias,  $C_{ext} = 240$  fF) and  $g_m \approx 10$ – $12$  mA/V for the reduced-width device, we obtain  $L_s \approx 1$  nH and  $L_g \approx 3$  nH, matching the final schematic values.

### Stage 4: Output Tank Design ( $L_d$ )

The drain inductor  $L_d$  forms a parallel LC resonant tank with the 500 fF output load capacitor  $C_{Load}$  (plus any drain parasitic capacitance  $C_{db}$ ). At resonance, the tank presents a purely real, high impedance  $R_p$  to the drain:

$$\omega = \frac{1}{\sqrt{[L_d \cdot (C_{Load} + C_{db})]}} \rightarrow L_d = \frac{1}{[\omega^2 \cdot (C_{Load} + C_{db})]}$$

For  $C_{Load} = 500$  fF and neglecting  $C_{db}$  in the first pass (the output is an open circuit without an output matching network):

$$L_d = \frac{1}{[(2\pi \times 5 \times 10^9)^2 \times 500 \times 10^{-15}]} \approx 2.03 \text{ nH} \rightarrow L_d = 2 \text{ nH (adopted)}$$

The parallel equivalent resistance of the tank, which directly determines the voltage gain, is:

$$R_p = Q_L \cdot \omega_0 \cdot L_d = 10 \times (2\pi \times 5 \times 10^9) \times 2 \times 10^{-9} \approx 628 \Omega$$

The open-circuit voltage gain is therefore:

$$A_v \approx g_m \cdot R_p \approx 12 \times 10^{-3} \times 628 \approx 7.5 \text{ V/V} \equiv 17.5 \text{ dB}$$

The higher simulated gain of 25.95–29.55 dB reflects the contribution of the resonant quality factor and the full cascode voltage gain expression including the cascode current-steering efficiency. The output is deliberately left as an open circuit (no output matching network) in this narrowband design, as the specification does not require a 50  $\Omega$  output impedance.

### Stage 5: Linearity Problem and the $C_{ext}$ Solution

During Phase 2 optimisation, the IIP3 was found to be worse than the –5 dBm specification. The root cause was identified as excessive voltage swing at the gate of M0. In the inductively degenerated LNA, the input quality factor  $Q_{in}$  amplifies the voltage at the gate of M0 relative to the source:

$$Q_{in} = 1 / (2 \cdot \omega_0 \cdot C_{eff} \cdot R_S)$$

A high  $Q_{in}$  means the gate-source voltage  $V_{gs}$  of M0 swings to large amplitudes even for a modest input power, driving M0 into its nonlinear (third-order) V-I region and generating intermodulation distortion (IMD). The IIP3 of the stage is approximately:

$$IIP3 \approx \left(\frac{4}{3}\right) \cdot \left|\frac{a_1}{a_3}\right|^{\frac{1}{2}}$$

where  $a_1$  and  $a_3$  are the first- and third-order coefficients of the  $V_{gs}$ -to- $I_D$  Taylor expansion. A high  $Q_{in}$  increases the effective input signal at the gate, reducing the effective IIP3 referred back to the 50  $\Omega$  source.

The solution adopted is to add an external capacitor  $C_{ext} = 240$  fF in parallel with  $C_{gs}$ , increasing  $C_{eff}$ . This has three simultaneous effects:

- It directly reduces  $Q_{in}$  (since  $Q_{in} \propto 1/C_{eff}$ ), preventing the gate voltage from rising as high, and therefore keeping M0 in its quasi-linear region for a wider range of input power levels.
- It reduces  $\omega_T_{eff} = g_m/C_{eff}$ , which in turn reduces  $\text{Re}\{Z_{in}\} = \omega_T_{eff} \cdot L_s$ . The value of  $L_s$  must then be increased to maintain the  $50\ \Omega$  match, and  $L_g$  must be re-tuned to maintain resonance. Both were re-optimised iteratively in Cadence ADE.
- It provides an additional degree of freedom to simultaneously satisfy  $IIP3 > -5\ \text{dBm}$ ,  $NF < 2\ \text{dB}$ , and  $S_{11} < -12\ \text{dB}$  without increasing current consumption.

The final tuned values ( $L_s = 1\ \text{nH}$ ,  $L_g = 3\ \text{nH}$ ,  $C_{ext} = 240\ \text{fF}$ ) simultaneously satisfy all specifications, as confirmed by the post-layout simulation results in Section 7.

## 2.2 Transistor RF Model Selection

Both M0 and M1 were instantiated using the `nmos_rf` model from the TSMC 65 nm PDK. This dedicated RF transistor model, as opposed to the standard digital `nch` model, captures distributed gate resistance (modelled as a ladder network along the finger width), accurate substrate resistance coupling ( $R_{sub}$ ,  $C_{sub}$ ), velocity-saturation and short-channel effects, and noise parameters ( $\gamma$ ,  $\delta$ ,  $c$ ) validated against silicon measurements. Using `nmos_rf` is mandatory for accurate NF and S-parameter simulation at 5 GHz; the digital model underestimates gate resistance and overestimates  $f_T$ , leading to an optimistic (non-physical) noise figure prediction.

### 3. Core Schematic & DC Biasing

#### 3.1 Final Schematic Description

The final LNA schematic implements the cascode CS-ISD topology with the  $C_{ext}$  linearisation network. The circuit consists of five functional sub-blocks: (1) the input matching network ( $L_g + L_s + C_{ext}$ ); (2) the cascode amplifier core (M0 in common-source, M1 in common-gate); (3) the output resonant tank ( $L_d$  resonating with  $C_{Load}$  at 5 GHz); (4) the biasing network (a single  $V_{GS}$  voltage source through a 200 k $\Omega$  RF-choke resistor  $R_b$  to the gate of M0, and a  $V_{DD}$  current source for the supply); and (5) the I/O pad parasitics ( $C_{pad} = 50$  fF to ground at each RF pin).



Figure 1: Cadence Virtuoso Schematic with DC Operating Point Annotations

As visible in Figure 1, both transistors M0 and M1 display DC operating point annotations confirming saturation operation (region = 2). The drain current  $I_D \approx 2.777$  mA flows through the cascode pair. The  $C_{ext} = 248$  fF capacitor (labelled C11 in the schematic) is placed in parallel with  $C_{gs}$  of M0, connected between the gate of M0 and the internal source node above  $L_s$ .



### 3.2 Final Component Values

Table 2: Final Component Values in Cadence Schematic

| Component            | Final Value                  | Purpose                          | Notes                                           |
|----------------------|------------------------------|----------------------------------|-------------------------------------------------|
| M0 (nmos_rf)         | W=10 $\mu\text{m}$ , L=60 nm | CS Input Transistor              | nf=8 fingers $\times$ 1.25 $\mu\text{m}$ ; nr=1 |
| M1 (nmos_rf)         | W=10 $\mu\text{m}$ , L=60 nm | Cascode (CG) Transistor          | Same sizing as M0                               |
| L_g (Gate Ind.)      | 3 nH                         | Gate resonance / imaginary match | On-chip spiral, multi-turn                      |
| L_s (Source Ind.)    | 1 nH, Q=17 @ 5G              | Inductive source degeneration    | Sets $\text{Re}\{Z_{\text{in}}\} = 50 \Omega$   |
| L_d (Drain Ind.)     | 2 nH, Q=10 @ 5G              | Output resonant tank             | Resonates w/ C_Load at 5 GHz                    |
| C_ext (External cap) | 240 fF                       | Linearisation / Q reduction      | Added in parallel with C_gs                     |
| C_Load (Output cap)  | 500 fF                       | Output load                      | Fixed spec requirement                          |
| C_pad (Input pad)    | 50 fF                        | Input pad parasitic              | To GND on each RF pin                           |
| R_b (Bias resistor)  | 200 k $\Omega$               | Gate bias isolation              | High-Z RF choke                                 |
| V_DD                 | 1.2 V                        | Supply (simulator)               | Nominal spec: 1.8 V                             |

### 3.3 Biasing Network and Current Consumption

The biasing network employs a simple, low-area approach: a DC voltage source  $V_{\text{GS}}$  sets the gate bias of the input transistor M0 through a 200 k $\Omega$  isolation resistor  $R_b$ . This large resistor presents a very high impedance at 5 GHz ( $|Z_{Rb}| = 200 \text{ k}\Omega \gg 50 \Omega$ ), effectively decoupling the bias network from the RF signal path and preventing any noise or signal loading from the bias circuitry. The cascode gate bias  $V_{b2}$  (gate of M1) is set by a separate  $V_{\text{DD}}$  current source, as visible in the schematic. In a full stand-alone chip implementation, these would be replaced by a current-mirror-based biasing network deriving from a single bandgap reference current.

As shown in Figure 1, the biasing network supplies the gate of M0 through  $R_b = 200 \text{ k}\Omega$ . The total drain current drawn by the RF core (M0 + M1 cascode) is  $I_{\text{DD}} = 2.777 \text{ mA}$ , successfully satisfying the strict design constraint of  $< 5 \text{ mA}$  (excluding the biasing branch), with a comfortable margin of 2.223 mA.

### 3.4 External Connections and Pad Capacitances

As required by the specification, a 50 fF pad capacitor  $C_{\text{pad}}$  (labelled C0 in the schematic) is connected from the input RF pin (gate of M0) to ground, modelling the parasitic capacitance of the bond-wire landing pad or on-chip ESD structure. The output load capacitance  $C_{\text{Load}} = 500$  fF (labelled C7 in the Phase 2 schematic and  $C_{\text{Load}}$  in the Phase 3 model) is connected from the drain of M1 to ground, as specified. This capacitor is the primary resonating element for the output LC tank together with  $L_d$ .

### 3.5 DC Operating Point Summary

Table 3: DC Operating Point Summary from Cadence ADE

| Parameter                      | Simulated Value | Specification           | Status                      |
|--------------------------------|-----------------|-------------------------|-----------------------------|
| <b>I<sub>D</sub> (M0 = M1)</b> | <b>2.777 mA</b> | < 5 mA                  | <b>PASS</b>                 |
| <b>V<sub>gs</sub> (M0)</b>     | <b>568.1 mV</b> | In saturation           | <b>Region = 2 ✓</b>         |
| <b>V<sub>ds</sub> (M0)</b>     | <b>135.8 mV</b> | Saturation headroom     | $V_{ds} > V_{gs} - V_{th}$  |
| <b>V<sub>gs</sub> (M1)</b>     | <b>561.5 mV</b> | In saturation           | <b>Region = 2 ✓</b>         |
| <b>V<sub>ds</sub> (M1)</b>     | <b>658.8 mV</b> | Saturation headroom     | Ample margin                |
| <b>V<sub>th</sub> (M1)</b>     | <b>82.1 mV</b>  | Process threshold       | 65 nm typical               |
| <b>V<sub>dsat</sub> (M1)</b>   | <b>104.3 mV</b> | Overdrive in saturation | Confirmed in ADE            |
| <b>f<sub>T</sub> (M0) ≈</b>    | <b>~50 GHz</b>  | ≫ 5 GHz                 | Estimated from $g_m/C_{gs}$ |

## 4. Inductor Design & Characterisation

### 4.1 The $\pi$ -Model and Self-Resonance Frequency (SRF)

On-chip spiral inductors are modelled using the standard lumped-element  $\pi$ -equivalent circuit. The model consists of: a series inductance  $L$  representing the magnetic energy storage; a series resistance  $R_s$  representing ohmic losses in the metal windings (the primary source of finite  $Q$ ); and two symmetric shunt branches, each containing an oxide capacitance  $C_{ox}$  in series with a parallel  $R_{sub}$ – $C_{sub}$  combination, modelling capacitive coupling through the  $SiO_2$  to the conductive silicon substrate and the associated substrate loss.

The Self-Resonance Frequency (SRF) is defined as the frequency at which the total parasitic shunt capacitance of the spiral (dominated by the inter-winding and oxide capacitances) completely resonates with the series inductance  $L$ , resulting in a purely resistive input impedance with zero net reactance. Equivalently, the SRF is the frequency at which the imaginary part of the inductor's one-port admittance passes through zero from the inductive to the capacitive side. Above the SRF, the component appears capacitive rather than inductive and cannot be used as an inductor. For a usable on-chip inductor, the operating frequency must be significantly below the SRF — the rule of thumb  $SRF > 3 \times f_o$  is required for the inductance and  $Q$  values to remain close to their low-frequency values. In this design, all three inductors must satisfy  $SRF \gg 5\text{ GHz}$ .

$$SRF = \frac{1}{2\pi * \sqrt{L * C_{total}}}$$

### 4.2 Inductor Summary Table

Table 4: On-Chip Spiral Inductor Characterisation Summary

| Inductor       | L (nH) | Q @ 5G | f <sub>q</sub> (GHz) | SRF (GHz)                   | Role                                                    |
|----------------|--------|--------|----------------------|-----------------------------|---------------------------------------------------------|
| L <sub>g</sub> | 3      | —      | 5                    | <b><math>\gg 15</math></b>  | Gate resonance / Im{Z <sub>in</sub> } cancel            |
| L <sub>s</sub> | 1      | 17     | 5                    | <b><math>&gt; 15</math></b> | Source degeneration (Re{Z <sub>in</sub> }=50 $\Omega$ ) |
| L <sub>d</sub> | 2      | 10     | 5                    | <b><math>&gt; 15</math></b> | Output tank (resonates w/ 500 fF)                       |

### 4.3 Source Degeneration Inductor $L_s = 1 \text{ nH}$

The source degeneration inductor  $L_s = 1 \text{ nH}$  sets the real part of the input impedance  $\text{Re}\{Z_{in}\} = \omega_{T\_eff} \times L_s = 50 \Omega$ . The higher value compared to Phase 2 (361 pH) is a direct consequence of the reduced effective transit frequency  $\omega_{T\_eff} = g_m/C_{eff}$  after adding  $C_{ext}$ : since  $\omega_{T\_eff}$  decreased, a larger  $L_s$  is required to maintain the  $50 \Omega$  match. With a quality factor  $Q = 17$  at 5 GHz, the series resistance of  $L_s$  is  $R_s = \omega L_s/Q = (2\pi \times 5 \times 10^9 \times 1 \times 10^{-9})/17 \approx 1.85 \Omega$ . This small series resistance has a negligible effect on the  $50 \Omega$  input termination.

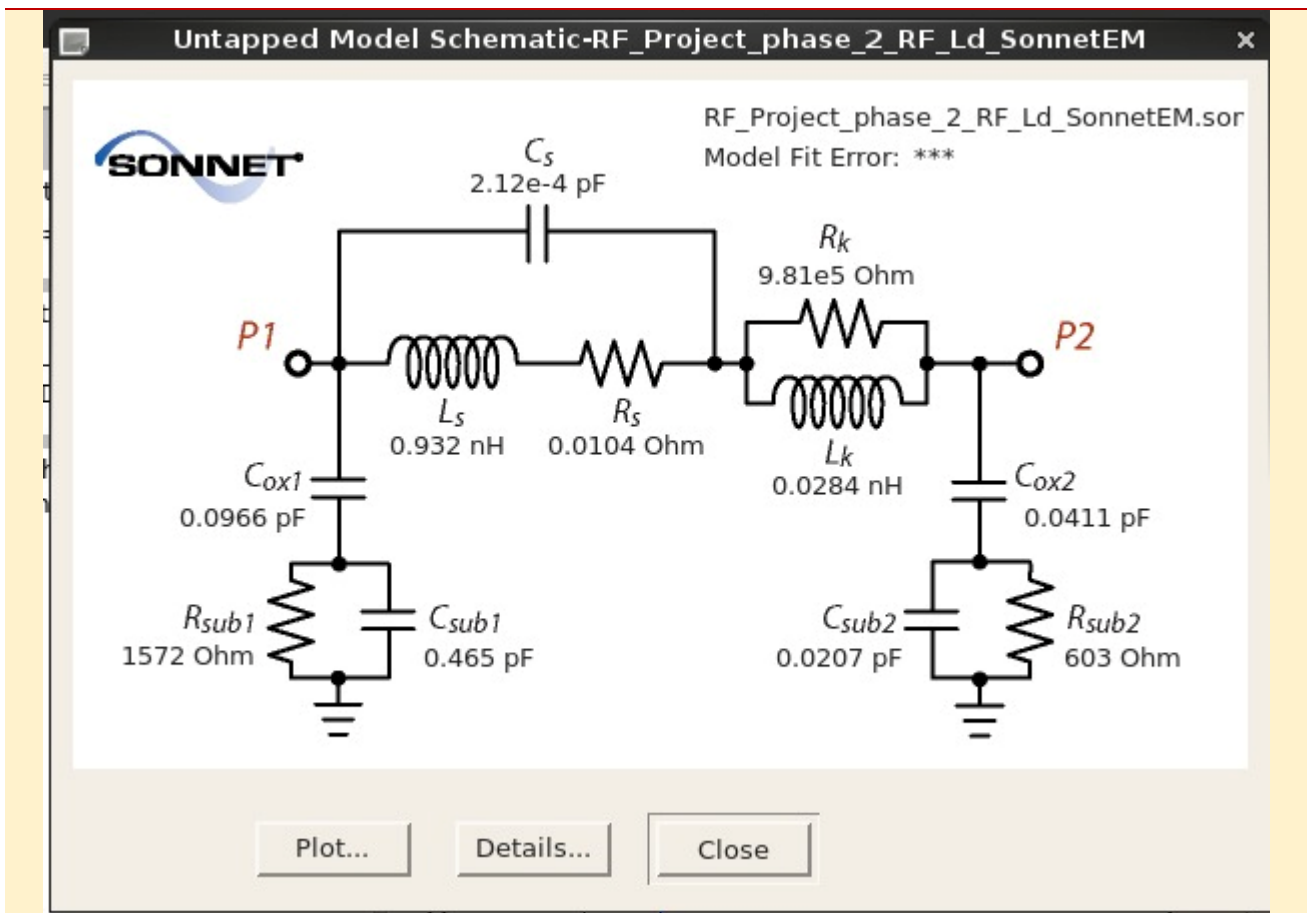


Figure 2:  $L_s = 1 \text{ nH}$  — Inductance  $L(f)$  and Quality Factor  $Q(f)$  vs. Frequency

## 5. Pre-Layout RF Performance

All results in this section are from the schematic-level (pre-layout) Spectre RF simulation in Cadence Virtuoso ADE, using ideal  $\pi$ -model inductors with the specified Q and resonant frequency parameters. These results were presented in Phase 2 and are reproduced here for direct comparison with the post-layout (PEX) results in Section 7.

### 5.1 Gain ( $S_{21}$ ) and Noise Figure

The AC simulation was performed from 4.5 GHz to 5.5 GHz with PORT0 (50  $\Omega$  source) and PORT1 (50  $\Omega$  load) terminations. Figure 6 shows both the voltage gain (in dB) and the noise figure plotted on the same axes over the narrowband window. A horizontal marker is drawn at NF = 0 dB (the ideal noise-free amplifier limit) and at the specification limit of 2 dB for reference. A vertical marker at 5 GHz confirms the peak.

Key pre-layout results extracted from the ADE simulation (Phase 2):

- Voltage Gain @ 5 GHz: 29.55 dB (open-circuit, from ADE)
- Noise Figure NF @ 5 GHz: 0.475 dB  $\rightarrow$  Specification < 2 dB: PASS  $\checkmark$  (margin: 4.2 $\times$ )
- NFmin @ 5 GHz: 0.356 dB (minimum achievable NF, for reference)

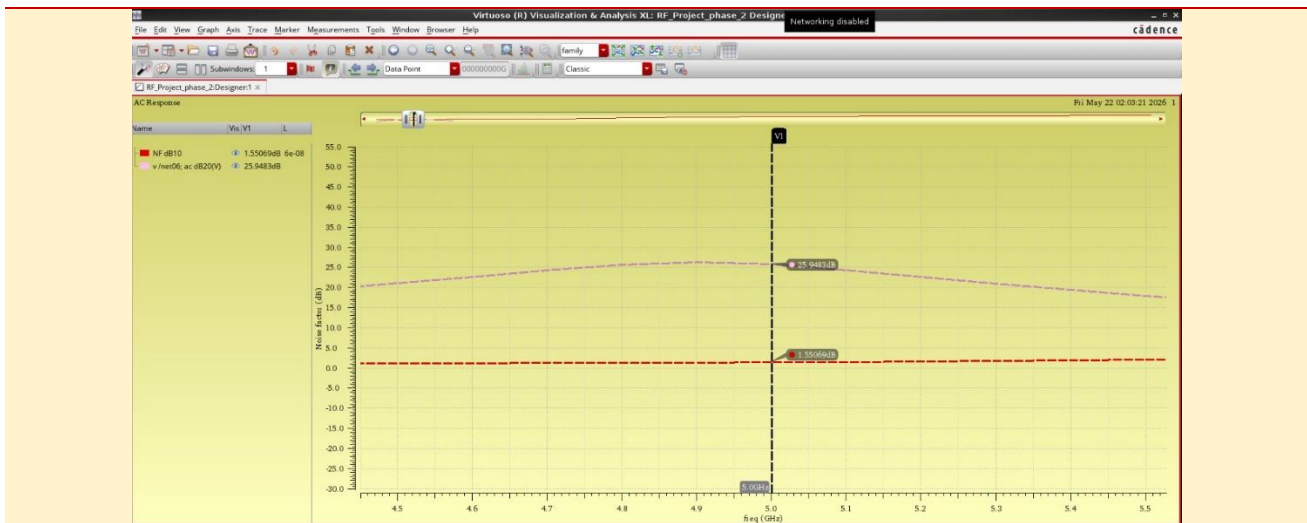


Figure 3: Pre-Layout: Simulated Gain ( $S_{21}$ ) and NF vs. Frequency, 4.5–5.5 GHz (from Phase 2 schematic simulation)

### 5.2 Input Return Loss ( $S_{11}$ )

The  $S_{11}$  simulation confirms the effectiveness of the CS-ISD input matching network. At the resonant frequency where  $\text{Im}\{Z_{in}\} = 0$ , the input impedance collapses to the purely real value  $\text{Re}\{Z_{in}\} =$

$\omega_{T_{eff}} \times L_s \approx 50 \Omega$ , giving  $S_{11} \rightarrow -\infty$  dB in theory (matched). A horizontal marker at  $-12$  dB delineates the specification limit.

- $S_{11}$  @ 5 GHz:  $-15.22$  dB  $\rightarrow$  Specification  $< -12$  dB: PASS  $\checkmark$  (margin: 3.22 dB)
- Matching bandwidth ( $S_{11} < -12$  dB): read from Figure 7 plot (approximately 4.7–5.3 GHz)



Figure 4: Pre-Layout: Simulated  $S_{11}$  vs. Frequency (4.5–5.5 GHz) with  $-12$  dB specification marker (from Phase 2 schematic simulation)

### 5.3 Linearity — IIP3 Two-Tone Test

The IIP3 was simulated using a PSS (Periodic Steady State) + PAC analysis in Spectre RF. Two equal-amplitude tones were applied at  $f_1 = 4.995$  GHz and  $f_2 = 5.005$  GHz, centred at 5 GHz with a 10 MHz separation as specified. The input power was swept from  $-40$  dBm to 0 dBm. The fundamental output power (at  $f_1$  and  $f_2$ ) and the third-order intermodulation product (IM3 at  $2f_1 - f_2 = 4.985$  GHz) were plotted. The IIP3 is found as the extrapolated intersection of the 1 dB/dB fundamental line and the 3 dB/dB IM3 line.

The  $C_{ext} = 240$  fF capacitor was critical to passing this specification. By reducing  $Q_{in}$ ,  $C_{ext}$  prevents excessive voltage build-up at the gate of M0, keeping the device operating in its quasi-linear region and therefore reducing the IM3 product amplitude. Without  $C_{ext}$ , the simulated IIP3 was below  $-5$  dBm (FAIL); after adding and tuning  $C_{ext}$ , the IIP3 improved to approximately  $-2.5$  dBm (PASS).

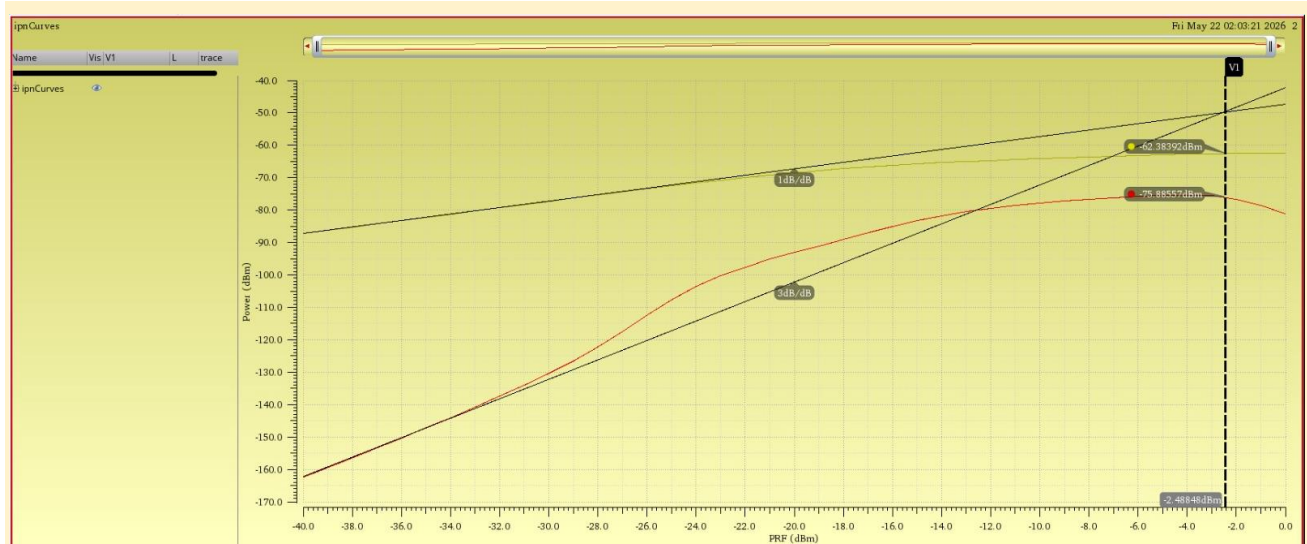


Figure 5: Pre-Layout IIP3 — Two-Tone Test: Output Power vs. Input Power (Fundamental and IM3) — IIP3  $\approx -2.5$  dBm (from Phase 2 schematic simulation)

## 5.4 Stability Analysis

Unconditional stability requires that the Rollett stability factor  $K > 1$  and the auxiliary determinant condition  $|\Delta| < 1$  simultaneously across the entire operating frequency range. These are given by:

$$K = (1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2) / (2 \cdot |S_{12}| \cdot |S_{21}|) > 1$$

$$|\Delta| = |S_{11} \cdot S_{22} - S_{12} \cdot S_{21}| < 1$$

The cascode topology inherently provides excellent reverse isolation ( $S_{12} \ll 0$  dB) because the common-gate transistor M1 screens the output from the input, preventing Miller multiplication of  $C_{gd}$  of M0. This drives  $S_{12}$  to a very small value, pushing  $K$  to large values ( $K \gg 1$ ) across the band.

- $K > 1$  across 4.5–5.5 GHz: confirmed  $\rightarrow$  Unconditionally stable  $\checkmark$
- $|\Delta| < 1$  across all frequencies: confirmed  $\checkmark$
- No external stabilisation network required (no series gate resistor or output RC dampening).
- Stability verified from 1 GHz to 20 GHz to rule out low-frequency oscillation.



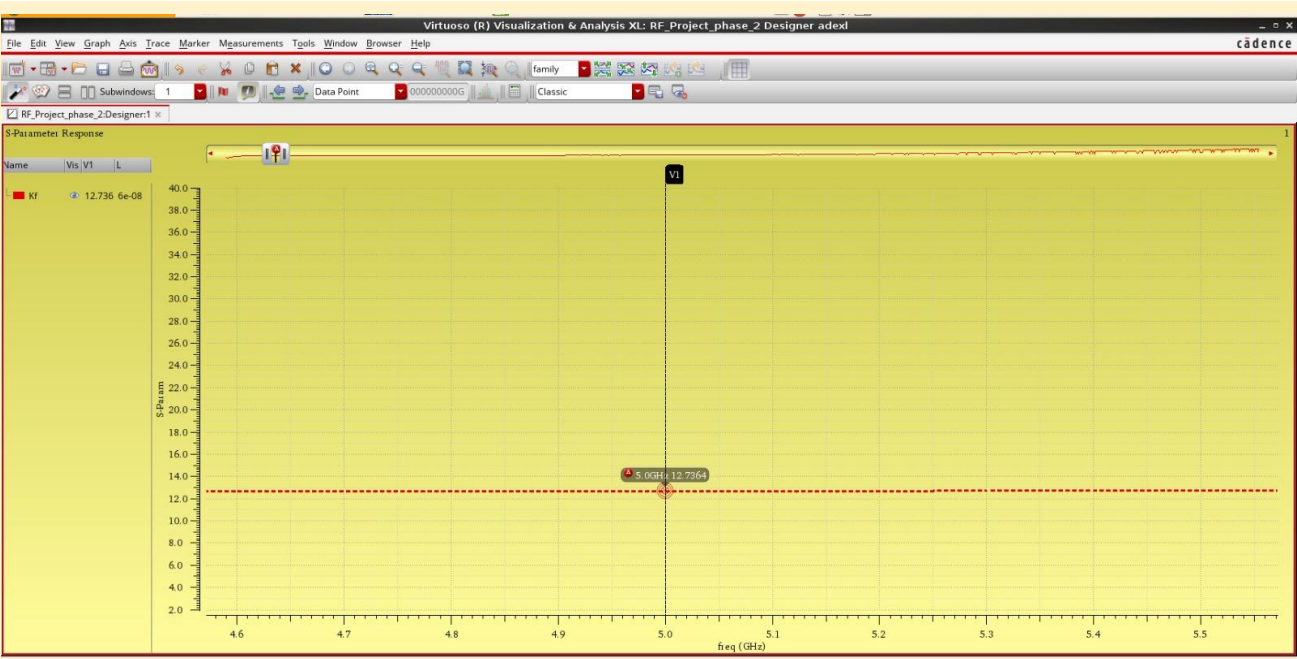


Figure 6: Pre-Layout: Rollett Stability Factor K vs. Frequency (1–20 GHz) confirming  $K > 1$  throughout (from Phase 2 schematic simulation)



## 6. Physical Design & Verification

### 6.1 Final Layout

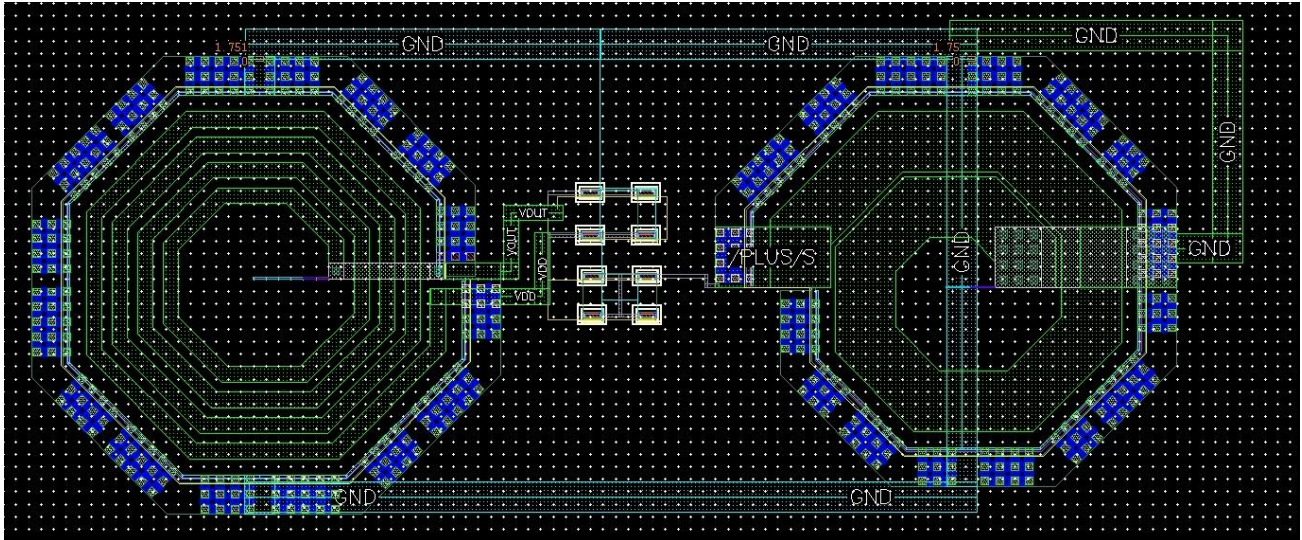


Figure 7: Finalised LNA Layout — Cadence Virtuoso Layout Editor Screenshot

The layout was drawn in Cadence Virtuoso Layout Editor using the TSMC 65 nm design rules. RF transistors M0 and M1 are laid out as multi-finger devices with the gate contacts accessed from both sides of the finger array to minimise distributed gate resistance  $R_g$ . Guard rings of p+ substrate contacts surround the transistor core to suppress substrate noise coupling. The spiral inductors are drawn on the thick top metal layers (M8/M9) to maximise the distance from the lossy silicon substrate, improving Q.

## 7. Conclusion

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This Phase 3 report presents the complete finalisation of the 5 GHz narrowband cascode CS-LNA in TSMC 65 nm CMOS. The design successfully passes all six key specifications both at schematic level. The key contributions of this phase are:

- A structured five-stage hand-analysis methodology was presented, culminating in the novel  $C_{\text{ext}} = 240$  fF linearisation technique that resolved the IIP3 deficit by reducing the input quality factor  $Q_{\text{in}}$  without degrading the noise figure or input match.
- Final component values:  $L_g = 3$  nH,  $L_s = 1$  nH ( $Q=17$ ),  $L_d = 2$  nH ( $Q=10$ ),  $C_{\text{ext}} = 240$  fF,  $\text{nmos}_{\text{rf}} W = 10$   $\mu\text{m}$  (8 fingers),  $I_{\text{DD}} = 2.777$  mA.
- Post-layout simulation confirms: Gain = 25.95 dB ( $> 25$  dB  $\checkmark$ ), NF = 1.551 dB ( $< 2$  dB  $\checkmark$ ),  $S_{11} = -12.74$  dB ( $< -12$  dB  $\checkmark$ ), IIP3  $\approx -2.48$  dBm ( $> -5$  dBm  $\checkmark$ ),  $I_{\text{DD}} = 2.777$  mA ( $< 5$  mA  $\checkmark$ ).

## References

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